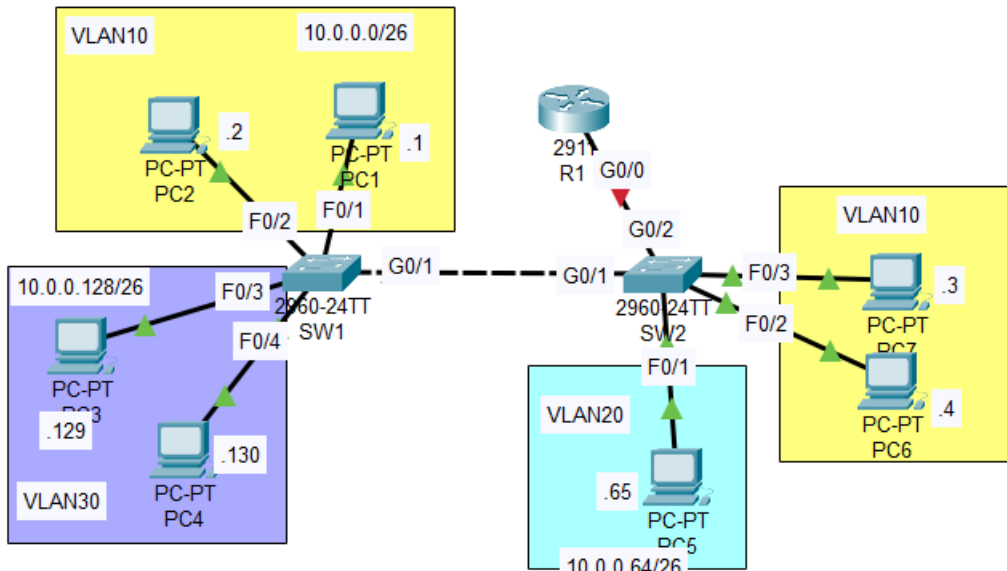




Topology

1. Configure the switch interfaces connected to PCs as access ports in the correct VLAN.

```
SW1>en
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int range f0/1-2
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
SW1(config-if-range)#int range f0/3-4
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 30
% Access VLAN does not exist. Creating vlan 30
SW2>en
SW2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#int f0/1
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
SW2(config-if)#int range f0/2-3
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
```

2. Configure the connection between SW1 and SW2 as a trunk, allowing only the necessary VLANs. Configure an unused VLAN as the native VLAN. Make sure all necessary VLANs exist on each switch.

```

SW1(config)#int g0/1
SW1(config-if)#switchport trunk encapsulation dot1q
^
% Invalid input detected at '^' marker.

SW1(config-if)#switchport mode trunk

SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW1(config-if)#int g0/1
SW1(config-if)#switchport trunk allowed vlan 10,30
SW1(config-if)#switchport trunk native vlan 1001
SW2(config-if)#int g0/2
SW2(config-if)#switchport mode trunk
SW2(config-if)#switchport trunk allowed vlan 10,20,30
SW2(config-if)#switchport trunk native vlan 1001
SW2(config)#vlan 30
SW2(config-vlan)#name VLAN0030
SW2(config-vlan)#show vlan brief

```

3. Configure the connection between SW2 and R1 using 'router on a stick'. Assign the last usable address of each subnet to R1's subinterfaces.

```

R1(config)#int g0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#interface g0/0.10
R1(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip address 10.0.0.62 255.255.255.192
R1(config-subif)#interface g0/0.20
R1(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

R1(config-subif)#encapsulation dot1q 20
R1(config-subif)#ip address 10.0.0.126 255.255.255.192
R1(config-subif)#interface g0/0.30
R1(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

R1(config-subif)#encapsulation dot1q 30
R1(config-subif)#ip address 10.0.0.190 255.255.255.192

```

4. Test connectivity by pinging between PCs. All PCs should be able to reach each other.